

# Skills Summary

## Factoring Cubes and Higher-Degree Expressions

*Use this reference while completing your worksheet or when studying.*

### Skill 1 — Sum and difference of cubes

**Patterns:**  $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$   
 $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

**Signs (SOAP):** the binomial keeps the **S**ame sign; the trinomial's middle term takes the **O**pposite sign; its last term is **A**lways **P**ositive.

Cubes on sight: 1, 8, 27, 64, 125, 216. The trinomial never factors again over the integers.

**Example:** Factor  $x^3 + 27$

**Step 1.** Two terms, both perfect cubes, so this is a cube pattern — not something to group or treat as a quadratic.

**Step 2.** Here  $a = x$  and  $b = 3$ . The sum pattern gives the binomial  $x + 3$ , then  $x^2 - 3x + 9$ .  
 $\Rightarrow (x + 3)(x^2 - 3x + 9)$

**Try it:** Factor  $x^3 - 64$

(check:  $(x + 4)(x^2 - 4x + 16)$ )

### Skill 2 — Factor by grouping

**When:** four terms. Split them into two pairs, pull the GCF out of each pair, and the two leftover binomials must *match* — that match is what you factor out. If they differ only in sign, take  $-1$  out of the second pair.

Then keep going: the remaining quadratic is often a difference of squares.

**Example:** Factor  $x^3 - 3x^2 - 4x + 12$

**Step 1.** Four terms with no factor common to all of them, so I try grouping in pairs.

**Step 2.**  $x^2(x - 3) - 4(x - 3) = (x - 3)(x^2 - 4)$ , and  $x^2 - 4$  still splits.

$\Rightarrow (x - 3)(x - 2)(x + 2)$

**Try it:** Factor  $x^3 + 5x^2 - x - 5$

(check:  $(1 + x)(1 - x)(5 + x)$ )

### Skill 3 — Quadratic in disguise

**When:** the exponents come in the pattern  $2n$  and  $n$ , as in  $x^4 - 10x^2 + 9$  (here  $n = 2$ ). Substitute  $u = x^n$ , factor the ordinary quadratic in  $u$ , then put  $x^n$  back.

**You are not finished there:** each bracket in  $x^2$  is usually a difference of squares and has to be split too.

**Example:** Factor  $x^4 - 10x^2 + 9$

**Step 1.** The exponents are four and two, so  $u = x^2$  turns this into a quadratic I already know how to factor.

**Step 2.**  $u^2 - 10u + 9 = (u - 1)(u - 9)$ , so back in  $x$  it is  $(x^2 - 1)(x^2 - 9)$  — and both brackets split.

$$\Rightarrow (x - 1)(x + 1)(x - 3)(x + 3)$$

**Try it:** Factor  $x^4 - 29x^2 + 100$

$(x + 5)(x - 5)(x + 2)(x - 2)$  :check

### Skill 4 — From factors to zeros and shape

**Rule:** once a polynomial is a product of factors, set each factor equal to zero — the zero-product property turns one hard equation into several easy ones.

A factor of  $x$  contributes the zero  $x = 0$ . A repeated factor makes the graph *touch* the axis; a single factor makes it *cross*, so the sign flips there.

**Example:** Real zeros of  $x^3 - x^2 - 6x$

**Step 1.** I want the zeros, and a factored form hands them over, so I factor completely first.

**Step 2.**  $x(x^2 - x - 6) = x(x - 3)(x + 2)$ , so each of the three factors gives one zero.

$$\Rightarrow x = -2, 0, 3$$

**Try it:** Real zeros of  $x^3 - 9x$

$x = 0, 3, -3$  :check